Scalability & Future Plan

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Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the Electric Vehicle Market Performance & Insights project can be expanded to support more EV records, additional manufacturers and countries, more frequent data refreshes, and richer analytical features. The plan focuses on improving data processing, dashboard responsiveness, security, and future business insights.

Current System Limitations

S.No.	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Dataset coverage may be limited to a specific time period, countries, or vehicle categories.	Trends and comparisons may not represent the full EV market.	**High**
2	Missing, inconsistent, or duplicated values may exist in sales, battery, range, or rating fields.	Can reduce the accuracy and reliability of KPIs and visualizations.	**High**
3	Large datasets and multiple dashboard filters may increase query and rendering time.	Slower analysis and less responsive dashboard experience.	**Medium**

Scalability Plan

S.No.	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Designed primarily for individual/student analysis.	Publish a shared dashboard with role-based access and optimize concurrent-user performance if adopted by more users.
2	Data Storage	Uses the available project dataset and defined tables.	Store raw and cleaned data in separate layers; add scheduled ingestion and archive historical snapshots.
3	Performance	Charts and SQL queries operate on the current dataset size.	Optimize SQL, filter early, use aggregated tables/materialized views where supported, and monitor query history.
4	Security	Access depends on the current workspace and account setup.	Apply least-privilege roles, protect credentials, and limit access to sensitive or licensed data.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Clean and validate current EV dataset; document data dictionary and KPI definitions.	Short term 1–2 weeks	More consistent and trustworthy reporting.
Phase 2	Add more years, countries, manufacturers, vehicle categories, and battery technology records.	Short term 2–4 weeks	Broader market coverage and improved comparisons.
Phase 3	Create scheduled data refreshes, optimize SQL, and add dashboard filters and summary KPIs.	Medium term 1–2 months	Fresher data and faster dashboard interaction.
Phase 4	Add forecasting, market-share trends, charging infrastructure, and emissions/sustainability indicators.	Long term 3–6 months	Deeper strategic insights and stronger support for planning.

Note: This is a proposed future plan for the individual EV analytics project. Timelines and upgrades are recommendations, not completed implementation claims.